গবেষণা, পরীক্ষণ ও কনসালটেনসি কেন্দ্র
পুর ও পরিবেশ কৌশল বিভাগ
শাহজালাল বিজ্ঞান ও প্রযুক্তি বিশ্ববিদ্যালয়
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**Ref. no: CRTC/CEE/G20220953/S-145**

# SUB-SOIL INVESTIGATION REPORT

## FOR THE PROPOSED
## Five Storied Residential Building

**AT VILL-SATHPUR, BARLEKHA, MOULVIBAZAR**
**DAG # 1171, 1177, 1178, 1179, KHA # 682, J.L- 114, MOUJA- MOHUBOND**

**CLIENT**

**MOHAMMED MUKSAR UDDIN**
**S/O- HAJI MOHAMMED JOYNAL UDDIN**

Verification Code: 6B6Hy6h8
Verify report at: crtc-cee-sust.com

**CONDUCTED BY**

CENTER FOR RESEARCH TESTING & CONSULTANCY
DEPARTMENT OF CIVIL & ENVIRONMENTAL ENGINEERING
SHAHJALAL UNIVERSITY OF SCIENCE & TECHNOLOGY, SYLHET

**SAMPLE COLLECTED BY**

**UNIFIED SOIL TEST & ENGINEERING**
Uttaran-27 West Sheikh ghat, Sylhet
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Introduction

A reasonably accurate conception about the sub-soil parameters of any project site is an essential prerequisite for proper planning and designing the foundation of the structure so that the same after its construction would remain safe and serviceable. A detailed study on the sub-surface elements in this regard would be necessary to select proper type of foundation.

**Objectives**

The purpose of sub-soil investigation program is to obtain the following information

1. To classify the formation of soil layers
2. To estimate the safe bearing capacities of the foundation at different soil layers
3. To suggest the safe and appropriate type of foundations.

**Scope of work**

Centre for research, testing and consultancy (CRTC), Department of Civil and Environmental Engineering, Shahjalal University of Science & Technology (SUST), Sylhet; execute only the laboratory tests and other computational works with the supplied disturbed and/or undisturbed samples to determine the geotechnical properties of the soil.

**Table – 1: Items of the supplied samples**

| No | Name | Bore hole no | Depth | Quantities |
|---|---|---|---|---|
| 01 | Disturbed sample | 01 | 0-50 ft | 10 |
| 02 | Disturbed sample | 02 | 0-60 ft | 12 |
| 03 | Disturbed sample | 03 | 0-50 ft | 10 |
| 04 | Disturbed sample | 04 | 0-50 ft | 10 |
| 05 | Disturbed sample | 05 | 0-50 ft | 10 |

**Laboratory Classifications**

The disturbed soil samples supplied by the client were thoroughly examined and classified roughly (color and type) and noted in accordance with bore hole (BH) and depth (Please see enclosed Tables).

**Laboratory Works & Analysis**

In addition to visual examination of the sample, Bearing Capacity of soil samples including skin friction & end bearing were calculated.

**Bearing capacity for shallow foundation**

The N value gives useful information with regards to consistency of cohesive soil and relative density of cohesion less soil. These relationships are given in the following Tables for cohesive and cohesion less soil, respectively.

**Table 2: Relation of N-value with the Consistency and the Unconfined Compressive Strength of Cohesive Soils (after Bowles 1997)**

| N-values | Consistency | Unconfined Compressive Strength ($q_u$) | |
|---|---|---|---|
| | | tsf | kPa |
| 0-2 | Very soft | 0 to 0.25 | 0 to 25 |
| 3-5 | Soft | 0.25 to 0.50 | 25 to 50 |
| 6-9 | Medium | 0.50 to 1 | 50 to 100 |
| 10-16 | Stiff | 1 to 2 | 100 to 200 |
| 17-30 | Very stiff | 2 to 4 | 200 to 400 |
| >30 | Hard | >4 | >400 |

**Table 3: N and ϕ Relation to Relative Density for Fine Sand (after Bowles 1997)**

| N values | Compactness | Relative Density, Dr % | ϕ (degree) | Unit Weight (pcf) |
|---|---|---|---|---|
| 1 – 2 | Very Loose | 0 – 15 | 26 – 28 | 70 – 100 |
| 3 – 6 | Loose | 15 – 35 | 28 – 30 | 90 – 115 |
| 7 – 15 | Medium | 35 – 65 | 30 – 34 | 110 – 130 |
| 16 – 30 | Dense | 65 – 85 | 33 – 38 | 110 – 140 |
| >30 | Very Dense | >85 | >50 | 130 – 150 |

**Table 4: N and ϕ Relation to Relative Density for Medium Sand (after Bowles 1997)**

| N values | Compactness | Relative Density, Dr % | ϕ (degree) | Unit Weight (pcf) |
|---|---|---|---|---|
| 2 – 3 | Very Loose | 0 – 15 | 27 – 28 | 70 – 100 |
| 4 – 7 | Loose | 15 – 35 | 30 – 32 | 90 – 115 |
| 8 – 20 | Medium | 35 – 65 | 32 – 36 | 110 – 130 |
| 21 – 40 | Dense | 65 – 85 | 36 – 42 | 110 – 140 |

**Table 5: N and ϕ Relation to Relative Density for Coarse Sand (after Bowles 1997)**

| N values | Compactness | Relative Density, Dr % | ϕ (degree) | Unit Weight (pcf) |
|---|---|---|---|---|
| 3 – 6 | Very Loose | 0 – 15 | 28 – 30 | 70 – 100 |
| 5 – 9 | Loose | 15 – 35 | 30 – 34 | 90 – 115 |
| 10 – 25 | Medium | 35 – 65 | 33 – 40 | 110 – 130 |
| 26 – 40 | Dense | 65 – 85 | 40 – 50 | 110 – 140 |
| >40 | Very Dense | >85 | >50 | 130 – 150 |

**Bearing capacity for deep foundation**

Foundations, preferably piles are considered to transmit heavy load from superstructure to deep and hard strata. As the formation of soil strata consists of cohesive and non-cohesive type, the evaluation criteria for the determination of skin friction and the end bearing capacities of piles also vary. Details of the evaluation are provided in the following sub-headings.

**Table 6: Formula Used to Determine Bearing Capacity of Soil**

| **Shallow Foundation** | |
|---|---|
| For Cohesive Soil | |
| $q_a = q_u + \gamma \frac{D}{3}$<br>$q_a = q_u$(Common Practice) | Where, $q_a$=Allowable bearing capacity, $q_u$= Unconfined Compressive Strength, $\gamma$= Unit Weight of soil, $D$= Depth of footing |
| For Cohesion less Soil | |
| $q_{ult} = 30N;$<br>$q_a = q_{ult}/FS$ | $q_{ult}$= Ultimate bearing capacity in kPa, $N$ = SPT Number, $FS$= Factor of Safety |
| **Source: Bowles, J. E., "Physical and Geotechnical Properties of Soils", 2nd Edition** | |
| **Deep Foundation** | |
| For Cohesive Soil | |
| $f_{su} = \alpha C_u$ ;<br>$q_{pu} = N_c C_u$ | $\alpha$ = Adhesion factor, $f_{su}$ = Ultimate skin friction, $C_u$ = Undrained Shear Strength, $q_{pu}$ = Ultimate end bearing of pile, $N_c$ = Bearing capacity factor for deep foundation=9 |
| For Cohesion less Soil | |
| $f_{su} = 4N/200\ tsf$ ;<br>$q_{pu} = 4N\ tsf$ | $f_{su}$ = Ultimate skin friction, $q_{pu}$ = Ultimate end bearing of pile, $N$ = SPT Number |
| **Source: Rajapakse RA (2008), "Geotechnical Engineering Calculations and Rules of Thumb"** | |

Based on the above formula, the important parameters for foundation design have been determined and are given in **Table 7**, **Table 8, Table 9, Table 10 and Table 11** respectively.

**Table 7**: The allowable bearing capacity for shallow foundation in Tsf (FS=2.5)**

| Bore Hole No. | Depth | Design N | Clay | | | | | | Sand |
|---|---|---|---|---|---|---|---|---|---|
| | | | Continuous | | | Square(Isolated) | | | |
| | (ft) | - | D/B=0.25 | D/B=0.5 | D/B=1 | D/B=0.25 | D/B=0.5 | D/B=1 | |
| BH01 | 2.5-5.0 | 4 | 0.510 | 0.537 | 0.585 | 0.603 | 0.651 | 0.708 | 0.550 |
| | 5.0-10 | 9 | 1.028 | 1.081 | 1.179 | 1.214 | 1.312 | 1.427 | 1.108 |
| BH02 | 2.5-5.0 | 4 | 0.487 | 0.512 | 0.559 | 0.575 | 0.622 | 0.676 | 0.525 |
| | 5.0-10 | 4 | 0.464 | 0.488 | 0.532 | 0.548 | 0.592 | 0.644 | 0.500 |
| BH03 | 2.5-5.0 | 4 | 0.487 | 0.512 | 0.559 | 0.575 | 0.622 | 0.676 | 0.525 |
| | 5.0-10 | 10 | 1.112 | 1.170 | 1.275 | 1.313 | 1.419 | 1.543 | 1.198 |
| BH04 | 2.5-5.0 | 10 | 1.137 | 1.196 | 1.303 | 1.343 | 1.450 | 1.578 | 1.225 |
| | 5.0-10 | 9 | 0.988 | 1.039 | 1.133 | 1.167 | 1.261 | 1.371 | 1.065 |
| BH05 | 2.5-5.0 | 3 | 0.371 | 0.390 | 0.426 | 0.438 | 0.474 | 0.515 | 0.400 |
| | 5.0-10 | 4 | 0.464 | 0.488 | 0.532 | 0.548 | 0.592 | 0.644 | 0.500 |
| BH06 | 2.5-5.0 | - | - | - | - | - | - | - | - |
| | 5.0-10 | - | - | - | - | - | - | - | - |
| BH07 | 2.5-5.0 | - | - | - | - | - | - | - | - |
| | 5.0-10 | - | - | - | - | - | - | - | - |
| BH08 | 2.5-5.0 | - | - | - | - | - | - | - | - |
| | 5.0-10 | - | - | - | - | - | - | - | - |
| BH09 | 2.5-5.0 | - | - | - | - | - | - | - | - |
| | 5.0-10 | - | - | - | - | - | - | - | - |
| BH10 | 2.5-5.0 | - | - | - | - | - | - | - | - |
| | 5.0-10 | - | - | - | - | - | - | - | - |

** Designer should consider both the calculated design N value and settlement behavior of the corresponding soil profile while selecting shallow footings

**Table 8: The allowable skin friction for precast driven piles in Tsf (FS=2.5)**

| Depth | Bore Hole | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| ft | BH01 | BH02 | BH03 | BH04 | BH05 | BH06 | BH07 | BH08 | BH09 | BH10 |
| 5 | 0.096 | 0.074 | 0.074 | 0.118 | 0.049 | - | - | - | - | - |
| 10 | 0.074 | 0.096 | 0.049 | 0.074 | 0.096 | - | - | - | - | - |
| 15 | 0.138 | 0.118 | 0.156 | 0.152 | 0.074 | - | - | - | - | - |
| 20 | 0.136 | 0.074 | 0.152 | 0.074 | 0.118 | - | - | - | - | - |
| 25 | 0.118 | 0.144 | 0.096 | 0.049 | 0.049 | - | - | - | - | - |
| 30 | 0.156 | 0.074 | 0.096 | 0.074 | 0.074 | - | - | - | - | - |
| 35 | 0.096 | 0.218 | 0.118 | 0.096 | 0.096 | - | - | - | - | - |
| 40 | 0.138 | 0.240 | 0.152 | 0.128 | 0.128 | - | - | - | - | - |
| 45 | 0.136 | 0.348 | 0.184 | 0.160 | 0.168 | - | - | - | - | - |
| 50 | 0.160 | 0.152 | 0.216 | 0.192 | 0.192 | - | - | - | - | - |
| 55 | - | 0.160 | - | - | - | - | - | - | - | - |
| 60 | - | 0.168 | - | - | - | - | - | - | - | - |
| 65 | - | - | - | - | - | - | - | - | - | - |
| 70 | - | - | - | - | - | - | - | - | - | - |
| 75 | - | - | - | - | - | - | - | - | - | - |
| 80 | - | - | - | - | - | - | - | - | - | - |
| 85 | - | - | - | - | - | - | - | - | - | - |
| 90 | - | - | - | - | - | - | - | - | - | - |
| 95 | - | - | - | - | - | - | - | - | - | - |
| 100 | - | - | - | - | - | - | - | - | - | - |

**Table 9: The allowable end bearing for precast driven piles in Tsf (FS=3.0)**

| Depth | Bore Hole | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| ft | BH01 | BH02 | BH03 | BH04 | BH05 | BH06 | BH07 | BH08 | BH09 | BH10 |
| 5 | 0.900 | 0.675 | 0.675 | 1.125 | 0.450 | - | - | - | - | - |
| 10 | 0.675 | 0.900 | 0.450 | 0.675 | 0.900 | - | - | - | - | - |
| 15 | 1.350 | 1.125 | 1.575 | 30.400 | 0.675 | - | - | - | - | - |
| 20 | 27.200 | 0.675 | 30.400 | 0.675 | 1.125 | - | - | - | - | - |
| 25 | 1.125 | 28.800 | 0.900 | 0.450 | 0.450 | - | - | - | - | - |
| 30 | 1.575 | 0.675 | 0.900 | 0.675 | 0.675 | - | - | - | - | - |
| 35 | 0.900 | 2.250 | 1.125 | 0.900 | 0.900 | - | - | - | - | - |
| 40 | 1.350 | 3.825 | 30.400 | 25.600 | 25.600 | - | - | - | - | - |
| 45 | 27.200 | 5.850 | 36.800 | 32.000 | 33.600 | - | - | - | - | - |
| 50 | 32.000 | 30.400 | 43.200 | 38.400 | 38.400 | - | - | - | - | - |
| 55 | - | 32.000 | - | - | - | - | - | - | - | - |
| 60 | - | 33.600 | - | - | - | - | - | - | - | - |
| 65 | - | - | - | - | - | - | - | - | - | - |
| 70 | - | - | - | - | - | - | - | - | - | - |
| 75 | - | - | - | - | - | - | - | - | - | - |
| 80 | - | - | - | - | - | - | - | - | - | - |
| 85 | - | - | - | - | - | - | - | - | - | - |
| 90 | - | - | - | - | - | - | - | - | - | - |
| 95 | - | - | - | - | - | - | - | - | - | - |
| 100 | - | - | - | - | - | - | - | - | - | - |

**Table 10: The allowable skin friction for Bored piles in Tsf (FS=2.5)**

| Depth | Bore Hole | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| ft | BH01 | BH02 | BH03 | BH04 | BH05 | BH06 | BH07 | BH08 | BH09 | BH10 |
| 5 | 0.062 | 0.048 | 0.048 | 0.077 | 0.032 | - | - | - | - | - |
| 10 | 0.048 | 0.062 | 0.032 | 0.048 | 0.062 | - | - | - | - | - |
| 15 | 0.090 | 0.077 | 0.101 | 0.099 | 0.048 | - | - | - | - | - |
| 20 | 0.088 | 0.048 | 0.099 | 0.048 | 0.077 | - | - | - | - | - |
| 25 | 0.077 | 0.094 | 0.062 | 0.032 | 0.032 | - | - | - | - | - |
| 30 | 0.101 | 0.048 | 0.062 | 0.048 | 0.048 | - | - | - | - | - |
| 35 | 0.062 | 0.141 | 0.077 | 0.062 | 0.062 | - | - | - | - | - |
| 40 | 0.090 | 0.156 | 0.099 | 0.083 | 0.083 | - | - | - | - | - |
| 45 | 0.088 | 0.226 | 0.120 | 0.104 | 0.109 | - | - | - | - | - |
| 50 | 0.104 | 0.099 | 0.140 | 0.125 | 0.125 | - | - | - | - | - |
| 55 | - | 0.104 | - | - | - | - | - | - | - | - |
| 60 | - | 0.109 | - | - | - | - | - | - | - | - |
| 65 | - | - | - | - | - | - | - | - | - | - |
| 70 | - | - | - | - | - | - | - | - | - | - |
| 75 | - | - | - | - | - | - | - | - | - | - |
| 80 | - | - | - | - | - | - | - | - | - | - |
| 85 | - | - | - | - | - | - | - | - | - | - |
| 90 | - | - | - | - | - | - | - | - | - | - |
| 95 | - | - | - | - | - | - | - | - | - | - |
| 100 | - | - | - | - | - | - | - | - | - | - |

**Table 11: The allowable end bearing for Bored piles in Tsf (FS=2.5)**

| Depth | Bore Hole | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| ft | BH01 | BH02 | BH03 | BH04 | BH05 | BH06 | BH07 | BH08 | BH09 | BH10 |
| 5 | 0.405 | 0.304 | 0.304 | 0.506 | 0.203 | - | - | - | - | - |
| 10 | 0.304 | 0.405 | 0.203 | 0.304 | 0.405 | - | - | - | - | - |
| 15 | 0.608 | 0.506 | 0.709 | 13.680 | 0.304 | - | - | - | - | - |
| 20 | 12.240 | 0.304 | 13.680 | 0.304 | 0.506 | - | - | - | - | - |
| 25 | 0.506 | 12.960 | 0.405 | 0.203 | 0.203 | - | - | - | - | - |
| 30 | 0.709 | 0.304 | 0.405 | 0.304 | 0.304 | - | - | - | - | - |
| 35 | 0.405 | 1.013 | 0.506 | 0.405 | 0.405 | - | - | - | - | - |
| 40 | 0.608 | 1.721 | 13.680 | 11.520 | 11.520 | - | - | - | - | - |
| 45 | 12.240 | 2.633 | 16.560 | 14.400 | 15.120 | - | - | - | - | - |
| 50 | 14.400 | 13.680 | 19.440 | 17.280 | 17.280 | - | - | - | - | - |
| 55 | - | 14.400 | - | - | - | - | - | - | - | - |
| 60 | - | 15.120 | - | - | - | - | - | - | - | - |
| 65 | - | - | - | - | - | - | - | - | - | - |
| 70 | - | - | - | - | - | - | - | - | - | - |
| 75 | - | - | - | - | - | - | - | - | - | - |
| 80 | - | - | - | - | - | - | - | - | - | - |
| 85 | - | - | - | - | - | - | - | - | - | - |
| 90 | - | - | - | - | - | - | - | - | - | - |
| 95 | - | - | - | - | - | - | - | - | - | - |
| 100 | - | - | - | - | - | - | - | - | - | - |

## CONCLUSION AND RECOMMENDATION

On the basis of analysis and information provided by the owner following recommendations are suggested for the construction of the proposed five storied residential building.

***Soil profile scenario***

Overall soil information of the investigated site is quite irregular in between the boreholes. Moreover, vertical inconsistency exists in the prevailing soil formation. In the boreholes 1, 2 3 and 4 clay soil of soft to very soft consistency has been observed up to 15-20 ft depth followed by dense sandy layer of about 5ft thickness. Then again, clay soil of soft to very soft consistency has been observed up to 35-45 ft depth followed by dense sandy layer extended up to 50 ft depth (except borehole 2 where it is extended up to 60 ft depth). In the borehole 5 clay soil of soft to very soft consistency has been observed up to 35 ft depth followed by dense sandy layer up to 50 ft depth.

***Foundation options***

In this situation, the foundation should be selected considering both the dimension and design load for the proposed structure. Since the whole site is dominated by cohesive soil with soft to very soft consistency in the top 35-45 ft depth with an intermediate single sandy layer, shallow foundation without ground improvement is not suitable for this type of structure. In this regard, the whole site in accordance with the borehole condition may be densified by sand drain or wick drain method. The depth of ground improvement should be selected by the foundation/design engineer considering both pressure influence zone and degree of densification required to minimize uneven settlement. After soil improvement, appropriate pattern of shallow foundation may be selected on the basis of degree of ground improvement. However, to overcome the difficulties associated with the ground improvement deep foundation may be adopted.

***Safety provisions***

Since all the boreholes of the proposed site are dominated with cohesive soil, it is reasonably vulnerable to differential settlement. Therefore, it is strongly recommended to check the settlement of the site considering building load while designing either shallow or deep foundation. Moreover, the earthquake vulnerability of the project area as well as the quality management of the construction work including workmanship should be taken into account for the safe design of shallow or deep foundation.

***Attachments***

The layout of the bore holes of the investigated site and the bore-log profiles with SPT values is shown in the Appendix. No further investigation was done by the client.

***Special notes***

These suggestions are absolutely tentative, and selection of type, depth and dimensions of foundation would be decided by the Consultant or Design Engineer. It is also to be noted that the above recommendation is restricted only to the existing soil condition of **50-60 ft** as client provided.

| Report Prepared By | Test Assisted By | Counter Signed By |
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